

Hospital Management System

Roadmap Review & Market Analysis — Conversation Record

A discussion between Suhas and Claude on the HMS project roadmap, the Indian healthcare-software market, ABDM / AB-PMJAY compliance, and go-to-market considerations.

SUHAS

Shared an 8-month Hospital Management System (HMS) project roadmap (single-tenant, FastAPI + React + Tauri, PostgreSQL) targeting small to mid-size clinics and hospitals in India, and asked for a senior developer's assessment: is the roadmap good or does it need changes? Also raised the idea of adding a patient-facing view where patients can learn about their disease and its treatment, and invited clarifying questions.

CLAUDE

Overall the roadmap is well-organized — the sequencing (schema → API → web → desktop → modules → deploy) is sound. But there are gaps worth fixing before writing code, because they are expensive to retrofit.

The biggest gap: no ABDM / ABHA support

ABDM (Ayushman Bharat Digital Mission) has moved from optional to effectively mandatory heading into 2026. State-level enforcement has begun — Bihar's directive to AB-PMJAY-empanelled private hospitals is live, and other states are expected to follow. It is increasingly required for NABH accreditation and state insurance empanelment. Practically the HMS needs to create/verify an ABHA ID inside the registration workflow and generate clinical records as HL7 FHIR R4 bundles, plus register the facility on the HFR and doctors on the HPR.

This is a day-one design decision, not a later feature. The roadmap stores medicines as free JSON and diagnosis as plain text; FHIR needs coded terminologies. Designing the schema for ABDM now is mostly free; adding it in year two means rewriting the data layer. Build the integration around Phase 6–7, but design the schema for it in Phase 1.

This is a CMS, not really an HMS

The mapped workflow (OPD → consult → lab → pharmacy → billing) is an outpatient/clinic flow. A 'hospital' usually implies IPD: bed/ward management, nursing notes, OT/ICU, admission-discharge-transfer. The schema has no admissions, beds, wards, or nursing tables. That is fine — but decide deliberately whether the target is clinics (current design fits) or hospitals (a large module is missing).

Tech stack — two concrete fixes

- **create-react-app is dead** — the React team deprecated it. Use Vite instead; axios, router, antd, and React Query all work identically.
- **The Tauri 'offline capability' is hand-waved** — wrapping React in Tauri gives a window, not offline. True offline from the server needs local data sync, which is a serious unscoped feature. Clarify what 'offline' means before promising it.

Testing is stacked at the end

Phase 7 (month 7–8) for all testing is the classic solo-dev trap. Write pytest tests alongside each endpoint in Phase 2; it costs ~10% early and saves weeks later.

DPDP + audit logs are under-weighted

For Indian healthcare these are requirements, not notes. Audit logging (who viewed/edited which record) should be a first-class table and middleware from Phase 1. Encryption at rest and TLS in transit should both be baseline — drop the 'or at minimum access-controlled' wording.

Timeline realism

Eight months solo for backend + web + desktop + pharmacy + billing + payment gateway + deployment + pilot is aggressive. If ABDM is in scope, add 3–6 months or use an integration layer rather than building FHIR from scratch.

On the patient-facing feature

Good instinct, but there is a big difference between two versions. The safe version: the patient sees their own doctor's diagnosis, prescription, lab results, and care instructions in plain language — essentially a patient portal, low-risk, and aligned with ABDM's PHR direction. Build this. The risky version: the system auto-generates explanations of the disease and treatment (e.g., LLM-written medical content) — this creates real medical liability and accuracy exposure. If done at all it must be doctor-reviewed, clearly educational not prescriptive, and never suggest changing treatment. Recommendation: start with the portal; if you want educational content, use a curated doctor-attached library, not free-form generation.

Questions raised

- Solo or team? Changes the timeline entirely.
- Clinics or actual hospitals with inpatient beds? Determines whether the IPD module is needed.
- Are target buyers AB-PMJAY empanelled? If yes, ABDM is not optional.
- Patient feature — portal, app, or printout? And is the info the doctor's own notes or auto-generated?
- Pricing / business model? Single-tenant on-prem means maintaining N separate installs forever — an operational cost that must be priced in.

SUHAS

- Team of 2 members; expected timeline at least 8 months.
- Currently focused on small-mid hospitals; may include clinics — needs to discuss with teammate.
- Doesn't know anything about AB-PMJAY — asked for a deeper explanation.
- Patient feature is both web and app login; nothing is auto-generated by AI.
- Didn't understand the 5th point (operational burden).
- Asked: is this a good project to work on, what is its value to society, and do the targeted end users really need it?

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AB-PMJAY explained

AB-PMJAY = Ayushman Bharat – Pradhan Mantri Jan Arogya Yojana, India's national health-insurance scheme launched in 2018 and run by the National Health Authority (NHA). It provides up to ₹5 lakh per family per year for secondary and tertiary hospitalisation, cashless, at empanelled public and private hospitals nationwide. Roughly the bottom 50% of the country qualifies (~12 crore families), and since 2024 it also covers everyone aged 70+ regardless of income via the Vay Vandana Card. It is very large — over 116 million admissions and ₹1.73 lakh crore of sanctioned treatments since launch — and as of mid-2026 it is operational across all 28 states and 8 union territories.

Why it matters to the software: a hospital must be empanelled to serve scheme patients cashlessly and claim reimbursement. To keep empanelment, the government is now pushing those hospitals to be ABDM-compliant. Bihar's 'integrate with ABDM or face de-empanelment' directive is already live; as of Feb 2026 the compliance umbrella covers 36,229 empanelled hospitals and explicitly extends to the HIS, LIMS, and pharmacy systems that power them. So if target hospitals are (or want to be) AB-PMJAY empanelled, the HMS must be ABDM-ready or those hospitals literally cannot buy it.

Point 5 restated — single-tenant operational burden

The roadmap chose single-tenant: every hospital gets its own copy on its own server (versus multi-tenant, one central cloud all hospitals log into). With single-tenant, signing 30 hospitals means 30 separate installations to maintain forever — every bug fix, patch, and update pushed to and tested on 30 servers, each on slightly different OS/hardware/network. When hospital #17's database crashes at 11pm, that is your team of two. This isn't wrong (hospitals like on-prem data), but the maintenance cost must be in the pricing and capacity plan. Two people can support maybe 10–20 installs before support eats all build time. Charge a recurring annual maintenance fee, build remote diagnostics and automated backups from day one, and standardize deployments.

Is this a good project? Does anyone need it?

Yes, it is worthwhile with real social value — but it is a crowded, mature market, so the edge won't be 'an HMS exists,' it will be something more specific.

The societal value is real: most small Indian clinics still run on paper — referral slips, handwritten prescriptions, lab reports in folders patients carry around. Digitizing reduces medication errors, prevents duplicate tests, speeds billing/registration, and (via the patient portal) lets patients own their health history.

Caveats: the market is crowded and mature (Lifemaan, MediXcel, and others serve hundreds of hospitals).

'Hospitals need an HMS' is true, but 'they need another generic HMS from a new 2-person team' is a harder sell — healthcare buyers are conservative. The real question is 'why you?' A small team needs a wedge: hyper-local presence and service; price + simplicity for one specific clinic type; or being ABDM-native from the ground up.

Recommendation: narrow the target before building; start with clinics/OPD (which the current schema fits) and add inpatient later. Above all, talk to 3–5 actual small hospitals before writing more code.

SUHAS

Clarified: this is not a learning or portfolio project — they are building it to sell to hospitals.

CLAUDE

That changes the read — selling software into hospitals is a different game from building it, and building is the easy 40%. Key things to internalize before spending 8 months:

Validate demand before you build. The most common way small teams waste a year is building something nobody buys. Find one design-partner hospital now — one that agrees to give feedback and ideally pay something to be your first customer. If you can't find even one willing to say 'build this and we'll use it,' that's critical information to have before month one. If you can, they will shape the product and become your reference customer.

The sales cycle is long and built on trust. Hospitals are conservative buyers handing over patient data. Their first questions: what if you lose our data, who do we call at 2am, will you exist in three years? Expect 3–9 months from first conversation to signed-and-paid, per hospital. Budget for that runway.

Once you sell it, you are legally liable. DPDP encryption, RBAC, audit trails, and ABDM-readiness stop being 'notes' and become baseline obligations. Selling an ABDM-incapable system to an empanelled hospital actively threatens their empanelment.

'Why you' is now existential. There are mature Indian HMS products from ~₹40K/month with references hospitals can call. Your business depends on a reason a hospital picks two unknowns over an established vendor: deep local service, radical simplicity/price for one clinic type, or genuinely ABDM-native. 'A solid general-purpose HMS' is not a wedge.

Support is a business commitment, not a Phase 7 task. Paying hospitals means nights/weekends/data-recovery obligations. Price it as a recurring maintenance contract; it caps how many installs two people can babysit.

The failure mode here isn't 'the code is bad' — it's 'we built the wrong thing for eight months, or built the right thing but can't sell or support it.' Those are business decisions being made right now, before any code exists. Questions: do you have any hospital already interested; are you full-time or alongside jobs/studies; and how will you make money (one-time licence, subscription, per-bed/per-doctor)?

SUHAS

Clarified they have not yet approached any doctor or hospital, and asked for an analysis of the current situation: is HMS really in high demand, and is there already software available for hospitals?

CLAUDE

Is HMS in high demand? Yes — genuinely.

- Market size/growth: the India HMS market is projected to grow from ~USD 12.4B (2025) to USD 25.7B (2031), ~12.8% CAGR; India is part of Asia Pacific, the fastest-growing region.
- Huge greenfield: in emerging markets like India, 60–70% of healthcare facilities still run on legacy or paper-based processes — most small hospitals haven't bought anything yet.
- Government tailwind: ABDM and AB-PMJAY actively push hospitals toward digitization, so the 'why digitize?' case is made for you.
- The gap is at your segment: small/mid hospitals repeatedly cite high upfront cost as the main barrier — precisely the affordable end you're aiming at.

So yes — small/mid Indian hospitals need this, many lack it, and the trend favors you.

Is there already software? Yes — a lot.

The market is crowded and mature, including at your target segment. Dozens of established Indian vendors already sell to clinics and small-mid hospitals — Lifemaan (328+ hospitals/clinics), MediXcel, MocDoc, Plus91, Birlamedisoft, ACGIL, Insta Health, Pappyjoe — plus enterprise players (Cerner/Oracle Health, Epic). Many already do OPD/IPD, billing, ABDM-readiness, cloud, and some are ahead with AI documentation and speech-to-prescription. Critically, domestic vendors have captured more than half of new installations in small-mid facilities specifically through lower pricing and local-language support — so 'cheaper and local' is already the incumbents' playbook.

What this means for two people

'High demand + lots of existing software' is the normal state of most good markets — it's validation, not a red light. But success has nothing to do with whether HMS is needed (it is) and everything to do with whether a specific hospital picks you over Lifemaan or MocDoc.

You cannot win by building 'a good HMS' — incumbents already have that plus references, support teams, and certification. You can only win by being the obvious choice for a narrow slice the big vendors serve badly: one specialty (dental, eye, IVF), one geography you can physically serve, or one painful job done exceptionally well. The narrower the better — a small team's only advantages are focus, speed, and personal service.

Recommendation

Don't start building yet. The make-or-break question is positioning, and that can only be answered by talking to hospitals, not by writing code:

- Talk to 5–10 small hospitals/clinics near you: what do you use today, what do you hate about it, why did/didn't you buy, are you AB-PMJAY empanelled, what would you pay?
- From those conversations, find the specific slice where people are unhappy or have nothing — that slice is your actual product.

- Only then decide what to build. If you can't find a slice where you'd clearly be someone's first choice, that's the most valuable thing you could learn — in a month instead of after eight.

Record of a working conversation. Market figures are drawn from third-party 2025-2026 industry sources and should be re-verified before use in any formal plan. Not legal or financial advice.